

KG

PORTFOLIO

Kristopher Gallow

AI-native operator and systems builder. Senior Capital Underwriter with a Process Automation focus at Toast, plus four LLM-powered terminals shipped through Ghost Strategies LLC.

[Portfolio HTML Link](#) [LinkedIn](#) [GitHub](#)

7+

Years in financial services

8

Production tools shipped

\$400M+Capital deployments
supported

FEATURED WORK

CM

COO/COLE Case Management Dashboard

Toast

Toast · 10-person underwriting team · 700+ active cases

Full-stack web app that replaced manual spreadsheet workflows. Real-time analytics scorecard (30-day, MoM, YTD, per-rep), Time-to-Decision distribution, Bizops pick-time attribution, fraud and TCA tabs, plus an Address Change tab with a five-status lifecycle.

Stack Apps Script, JavaScript, Tailwind, Chart.js (custom plugin) **Users** 10

ET

Email Template Automation Suite

Toast

Toast · 14-person Capital Underwriting and Change of Ownership · 11 phases

Cross-platform desktop app that cut template retrieval from 30 seconds to under 3 seconds. Zero AI tokens consumed, zero customer data exposure by design. Includes a Salesforce-embedded NOIA generator concept and a stakeholder presentation.

Stack Python, Flask, PowerShell, AppleScript, PyInstaller, pystray **Platforms** macOS + Windows

FV

Funding Verification Tool

Toast

Toast · Capital Underwriting · 20-35 minute workflow cut to 40 seconds

Single-file HTML application that replaced a fragile multi-source VLOOKUP chain in Google Sheets. In-browser four-source JOIN engine with processor-specific match logic, manual overrides, audit labels, and a template-based 4-sheet XLSX export that preserves the master sheet's conditional formatting.

Stack Vanilla JavaScript, SheetJS, custom RFC 4180 parser **Sources** 4-way JOIN across 9 tabs

CX

CSV Column Extractor

Toast

Toast · Bizops · 9-minute workflow cut to under 55 seconds

Single-file HTML tool for non-technical operations users. Streams 1M+ row spreadsheets via parallel Web Workers, with a pre-read and pre-warm pipeline that takes the full open-to-paste workflow from approximately nine minutes to under 55 seconds.

Stack Vanilla JavaScript, Web Workers, Papa Parse, SheetJS, Promise.all **Scale** 1M+ rows/file

GS

Ghost Strategies Terminal Suite

Independent

Ghost Strategies LLC · LLM-powered financial tooling

Four production terminals built with Anthropic's Claude: Wealth Compass (client-side financial planning), Training Terminal (Python, SQL, JavaScript, and Rust learning terminal), Garage Terminal (asset tracking on FastAPI), and Economic Indicators Terminal (BLS, BEA, FRED integration).

Stack Python, FastAPI, JavaScript, Pyodide, sql.js

Discipline I own the logic, the LLM writes the code, I validate every output



TOAST INC · SENIOR CAPITAL UNDERWRITER, PROCESS AUTOMATION

Toast Work

Four production tools built for the Capital Underwriting and Change of Ownership teams. Each replaced a fragile or manual workflow with a self-contained, browser-based system.

COO/COLE Case Management Dashboard

Full-stack web app · 10-person underwriting team · 700+ active cases

A web application built on top of a live Google Sheet that replaced manual spreadsheet workflows for case tracking, analytics, and address-change requests. The Cases tab tracks the full lifecycle of underwriting cases, the Address Change tab manages a parallel five-status workflow on its own sheet, and the Analytics tab is an operational scorecard with team-level and individual-rep views.

MOCK VIEW

● ● ● Case Management Dashboard · Cases (47 active · 3 picked today)

47

Active Cases

12

Completed (7d)

4.2

Avg Days TTC

98%

SLA Met

CASE #	RESTAURANT	SOURCE	STATUS	DAYS
C-1042	[Redacted]	Bizops	In Review	2
C-1041	[Redacted]	Collections	Picked	1
C-1040	[Redacted]	Direct	Completed	3
C-1039	[Redacted]	Bizops	In Review	5

KEY FEATURES

- Live case table backed by Google Sheets with inline edit drawer, add and delete modals, and a Settings panel for managers to self-serve configuration changes
- Address Change tab on a separate sheet with five-status lifecycle (Pending, Submitted to Risk, Completed, Expired, Canceled), independent filter bar, and automatic date stamping for creation and completion
- Analytics tab with 30-day, MoM, and YTD breakdowns; Time-to-Decision distribution buckets; and a Bizops pick-time attribution chart that correctly separates queue-wait time from rep turnaround time
- Custom Chart.js plugin (barTotalsPlugin) for stacked-bar totals and individual bar labels in two render modes
- Concurrency protection via LockService on case creation, deletion, and write operations
- Idle-refresh banner after 30 minutes of inactivity and a version-update notification system that polls the deployed app version every 10 minutes

STACK

Google Apps Script

HTML

CSS

JavaScript

Tailwind CSS

Chart.js v4

Google Sheets

Google OAuth

Email Template Automation Suite

11 phases · cross-platform desktop app · 14-person team · 30s to under 3s

A cross-platform desktop application that ships 20 pre-loaded email templates with rich-text formatting preserved on paste. Grew over 11 phases from a CLI tool to a full Flask web app, then to a PyInstaller-bundled native application with a system tray icon, and finally into a Salesforce-embedded NOIA generator concept presented to engineering stakeholders.

MOCK VIEW

● ● ● Email Templates · 20 templates · 14-person team

Capital Underwriting

Received Response · Template ready · ● Copied

NOIA

NOIA MCA · {date} · {restaurant_name} · {source_of_debt}

Change of Ownership

COO DS Sent · {date} · {restaurant_name}

Retrieval: avg 2.1s · Tokens consumed: 0 · Customer data sent to model: 0

PHASE PROGRESSION

- Phases 1 to 3: CLI tool, cross-platform GUI launcher, then a full Flask web app
- Phase 4: Dark mode, brand logo, and accordion drawer system for the editor
- Phase 5: PyInstaller-bundled standalone executable with platform-appropriate data directories (macOS app support, Windows AppData), port discovery to dodge AirPlay on macOS, and a system tray icon for clean exits
- Phase 6: Salesforce-embedded NOIA template generator concept, with function-based template rendering (eight templates, conditional logic for variable-length entity lists)
- Phases 7 to 9: Placeholder fill modal with confirmation toasts, multi-line field support, two-panel preview modal, and product demo enhancements
- Phase 10: Template-sync bug fix (silent failure mode on every launch caught and resolved)
- Phase 11: Stakeholder presentation HTML with Apps Script architecture diagrams and a NOIA automation proposal for engineering review

ENGINEERING DECISIONS

- Zero AI tokens consumed by design: templates use placeholders that the user fills locally, so no customer data ever leaves the machine
- Cross-platform clipboard formatting: macOS uses AppleScript with the «class HTML» format; Windows uses CF_HTML byte-offset headers with precise UTF-8 byte counting
- 15KB Flask app deliverable instead of a 150 to 200MB Electron bundle, deliberately avoiding the Apple Developer Program signing requirement

- Per-machine identity: no authentication needed, each install is local-only

STACK

Python

Flask

Bash

PowerShell

JavaScript

AppleScript

Quill.js

PyInstaller

pystray

Pillow

Funding Verification Tool

Single-file HTML · 20-35 minute workflow cut to 40 seconds

A single-file HTML application that replaced a fragile VLOOKUP chain in Google Sheets used to verify merchant banking data for capital deployments. Performs an in-browser JOIN across four sources (New Funding file, Snowflake export, Original CSV funding records, and DSL bank files), produces match results with processor-specific logic, supports real-time manual overrides, and exports a multi-sheet XLSX workbook from a bundled template that preserves the master sheet's conditional formatting. The tool grew to 9 tabs, including a TCA Check tab that auto-flags Toast Checking accounts and matches Slack-pasted GUIDs against an auto-detected canonical GUID column, and an External MCA Refi tab that tracks refi lineage between loans. The full funding verification workflow used to take 20 to 35 minutes per batch; the latest measured time (June 2026) clocked it at 40 seconds end-to-end.

BEFORE AND AFTER

BEFORE · 20-35 MINUTES

- 1 Manually maintain VLOOKUP chains across four Google Sheets
- 2 #N/A errors propagate when Snowflake data is delayed, no path to manual fix
- 3 No audit trail for manual SPA verifications or Toast Checking accounts
- 4 WORLDPAY and CHASE rules embedded in formulas, easily broken

AFTER · 40 SECONDS

- 1 Drag-drop four files, JOIN runs in milliseconds across 9 tabs
- 2 Manual Overrides tab surfaces Check Data rows, recompute on entry
- 3 Toast Checking and Verified in Merchant labels persist across re-uploads
- 4 Processor logic in code, 4-sheet XLSX export preserves master sheet formatting

MOCK VIEW

● ● ● Funding Verification · Ops SPA Deals (12 rows)

EXT ID	PROC	MATCH R	MATCH A	LABEL
EXT-8821	WORLDPAY	YES	YES	Toast Checking
EXT-8820	CHASE	YES	Check Data	Pending
EXT-8819	WORLDPAY	NO	YES	Verified
EXT-8818	CHASE	YES	YES	Verified

KEY ENGINEERING DECISIONS

- Custom RFC 4180 CSV parser written from scratch to replace SheetJS's unreliable CSV mode (BOM stripping, mixed line endings, embedded quotes)
- MID normalization for JavaScript scientific notation: Snowflake XLSX exports return Merchant IDs as 1.23E+9, breaking direct string compares
- Hash-join via Map objects: equivalent to NewFunding LEFT JOIN Snowflake LEFT JOIN OriginalCsv LEFT JOIN DSL, executes in milliseconds for hundreds of thousands of rows
- Persistent DSL Worker Pool (size 3): operators routinely drop 4 to 10 large DSL files at once, so XLSX parsing runs through a fixed pool of three Web Workers fed by a queue and dispatcher, with zero-copy ArrayBuffer transfer and workers reused across drops rather than re-spawned
- Processor-specific match logic isolated in code: WORLDPAY requires exact DSL-to-file match; CHASE requires file value to differ from a known placeholder
- Real-time override injection: when a Snowflake lookup misses, manual GUID/MID/Proc Partner entry constructs a synthetic Snowflake-like object and the JOIN re-runs without a page refresh
- Labels and overrides keyed to External ID, survive re-uploads of any source file within the same session
- Template-based XLSX export: bundles the master Google Sheet's exact conditional formatting rules in a base64 template and edits the template XML at runtime via JSZip to inject the live data, preserving every YES/NO/Check Data cell fill from the original
- TCA Check subsystem: auto-detects the canonical Toast GUID column per source using a UUID-density times uniqueness score, auto-flags Toast Checking accounts from the Original CSV, and matches Slack-pasted GUIDs against in-batch records with per-loan and per-merchant dedup modes
- Refi lineage tracking: the External MCA Refi tab links each refi loan to its original (matching the External ID minus its suffix) and highlights the original loans magenta in the XLSX export
- Multi-sheet audit export: one workbook with the Ops sheet plus additional audit sheets, each filename-stamped MMDDYYYY-Funding Verification Tool.xlsx
- 9-tab structure: Ops SPA Deals (primary output with stat-pill filter shortcuts), four source upload tabs, TCA Check, Refi, External MCA Refi, and Manual Overrides
- Ships with two companion artifacts: a 13-page Standard Operating Procedure (built with a python-docx script, 11 embedded screenshots, data-flow diagram, troubleshooting guide, and a domain glossary), and a standalone Funding File Validator that round-trip-checks the exported file against the source before handoff

STACK

Vanilla JavaScript

SheetJS 0.18.5

JSZip 3.10.1

Custom RFC 4180 parser

Web Worker pool

Web FileReader API

Base64 template embedding

Clipboard API

CSV Column Extractor

Single-file HTML · 9-minute workflow cut to under 55 seconds

A single-file HTML dashboard for combining up to five large spreadsheet files, each potentially exceeding one million rows, and extracting three specific columns for paste into a destination Google Sheet. Iterative performance engineering took the full open-to-paste workflow on three 25MB XLSX files from approximately nine minutes (manual baseline) to under 55 seconds.

BEFORE AND AFTER

BEFORE · ~9 MINUTES

- 1 Open three files sequentially in Excel or Sheets
- 2 Manually select and copy three columns from each file
- 3 Paste each chunk into the destination sheet, watch for cell-limit failures
- 4 Retry failed pastes after Google Sheets silently truncates

AFTER · UNDER 55 SECONDS

- 1 Drag-drop all three files into the tool
- 2 Pre-read and pre-warm pipeline starts in the background
- 3 Auto-parse fires 400ms after the last file, all three parse in parallel
- 4 Copy each column with one click, live cell-limit warnings prevent silent failures

MOCK VIEW

● ● ● CSV Column Extractor · 3 files queued · auto-parse in 0.4s

847,302

Total Rows

3

Files Parsed

~22s

Parse Time

3.9M

Cells / Chunk

COLUMN	ROWS	CELL ESTIMATE	ACTION
Column B (MID)	847,302	22.0M cells	Chunk Required
Column L (R&T)	847,302	22.0M cells	Chunk Required
Column M (DDA)	847,302	22.0M cells	Chunk Required

Chunk size 150K rows · 6 chunks per column · safely under 10M cell limit

PERFORMANCE PIPELINE

- Parallel parsing via Promise.all: all files parse simultaneously in independent workers with local accumulators, merged in original file order after all Promises resolve

- Pre-read FileReader cache: readAsArrayBuffer fires the moment a file enters the queue, not when Parse is clicked, eliminating I/O wait from the critical path
- Pre-warmed worker pool: a Web Worker is spawned per XLSX file on drop, compiling SheetJS (about 500KB) inside the worker during the user's file-review window
- Auto-parse with 400ms debounce: no Parse button click required, handles single-file and multi-file drops uniformly, guards against re-triggering once results are displayed
- Selective XLSX cell extraction: the worker constructs only three required cell addresses per row (B+n, L+n, M+n) instead of allocating the full 2D array via sheet_to_json
- Batch and chunk sizes increased 4x: worker message batches went from 25K to 100K rows; CSV chunk size went from 5MB to 20MB, reducing postMessage round-trips proportionally

BASE ENGINEERING DECISIONS

- Papa Parse with worker: true for CSV streaming; hand-written Blob Worker for SheetJS XLSX parsing to keep the UI responsive during 1M+ row binary parses
- ArrayBuffer transferred via postMessage transferables to eliminate the memory cost of a second copy on large files
- 150K-row clipboard chunks: 150K rows times 26 default Google Sheets columns equals 3.9M cells, safely under the 10M limit
- Push-in-loop array construction instead of spread or push.apply, avoiding call-stack overflow on arrays exceeding 125K elements
- Three primary copy buttons (one per column) because the destination sheet uses non-adjacent columns; combined tab-separated copy preserved as a secondary option

STACK

Vanilla JavaScript

Web Workers

Papa Parse 5.4.1

SheetJS 0.18.5

Blob Workers

Promise.all parallelism

FileReader API

Clipboard API



GHOST STRATEGIES LLC · INDEPENDENT

Ghost Strategies

Four LLM-powered financial tools built independently. Every project ships with the same discipline: I own the logic and architecture, the LLM writes the code, I validate every output, and I only prompt with test or sample data so no proprietary information ever enters the model.

Wealth Compass Terminal

Client-side financial planning dashboard

A client-facing financial planning dashboard that parses a 13-sheet workbook with 755 formulas entirely client-side. All data stays on the user's device. No backend, no data transmission, no model exposure to client information.

STACK

JavaScript

HTML

SheetJS

Custom formula engine

Training Terminal

Gamified four-language coding terminal · Python, SQL, JavaScript, Rust

A single-file, browser-based coding terminal (xterm.js) for practicing Python, SQL, JavaScript, and Rust. Runs real Python through Pyodide, real SQL through sql.js against a seeded relational database, and grades JavaScript inside a network- and storage-stripped Web Worker sandbox; Rust is graded by a literal-preserving structural matcher that accepts equivalent syntax. 200 curated questions — 50 per track across five gated tiers (Introductory, Amateur, Intermediate, Experienced, Master), 10 questions per gate — with localStorage-persisted progression and live tier and engine-status KPIs.

WHAT'S UNIQUE

- Four grading paths in one static file: Pyodide (Python), sql.js on a seeded auto-rebuilt database (SQL), an isolated Web Worker with network and storage APIs removed (JavaScript), and a structural normalizer with a syntax-equivalence system (Rust)
- Real xterm.js REPL: a command bar (start, hint, concepts, cheatsheet, examples, schema, reset, export/import), multi-line editing, command history, and emacs-style key bindings
- Learning layer baked in — per-tier concepts auto-surface on first entry, a toggleable cheatsheet panel, worked examples, and per-question hints
- 200 questions reviewed and calibrated by concept and difficulty; 10-question tier gates force mastery before the next rank unlocks, all persisted with no backend

STACK

Vanilla JavaScript

xterm.js

Pyodide

sql.js

Web Workers

localStorage

Garage Terminal

Multi-asset tracker · Python FastAPI backend

A personal asset tracking terminal with a Python FastAPI backend and an automated data pipeline using GitHub Actions for weekly serverless CI/CD workflows. Pulls valuation data, refreshes the dashboard, and exposes a clean read API.

STACK

Python

FastAPI

GitHub Actions

JavaScript

Serverless CI/CD

Economic Indicators Terminal

BLS, BEA, FRED API integration

A macroeconomic data terminal that integrates with three public US government economic data sources (BLS, BEA, FRED) via a Python proxy. Surfaces unified indicators (inflation, employment, GDP, regional metrics) in a single dashboard.

STACK

Python

BLS API

BEA API

FRED API

Proxy server

AI-Assisted Financial Modeling Suite

Dynamic financial engines built with Claude

A separate workstream under Ghost Strategies focused on financial modeling rather than terminal apps. Includes a probability-weighted valuation model and a \$3M TRS retirement planner. I own the financial logic and architecture while directing Claude to generate the code. Every output is validated against standard financial formulas before being trusted, and I only prompt with test or sample data to keep any real client information out of the model.



ABOUT

Background and operating model

How I work and what I'm looking for.

I'm a Senior Capital Underwriter at Toast, where I support over \$400M in active capital deployments and operate inside a fintech that scaled from \$100M to over \$2B in annual volume during my tenure. My day-to-day is capital decisions, customer health monitoring, and scenario planning, but my distinguishing work is building the systems and tools that operate alongside that core function.

I'm part of Toast's Claude Code and Cowork rollout to a select group of process automation specialists. I built a multi-agent system with an orchestrator that routes work to specialized sub-agents (workflow automation, code audit, technical project management, devops, technical writing), each with its own system prompt, scoped tool access, and isolated context. The same architecture is replicated as Skills in Cowork.

Outside of Toast, I run Ghost Strategies LLC, where I ship LLM-powered financial tools. My operating discipline across every project is consistent: I own the logic and architecture, the LLM writes the code, I validate every output before I trust it, and I never feed proprietary data to a model.

I'm based in San Antonio, Texas, and I work remote-only. I'm targeting Senior Manager-track or Senior IC roles in strategy, operations, risk, or AI-process work at companies where the AI-native operating model is the default, not the exception.

CORE STACK

SQL (Snowflake)

Salesforce

Python

JavaScript

Google Apps Script

Hex

Excel and Google Sheets

Claude Code

Cowork

Multi-Agent Systems

Flask

FastAPI

Chart.js



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